Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Constant acceleration

Meaning in my words How would you explain constant acceleration to a friend?	**How can you tell?** The velocities are 1, 3 and 5 m/s one second apart. What repeats?
Example or sketch If those gains continue, what velocity would you write for the next second?	**Non-example and why** Someone adds 2 m to position every second and calls that speeding up. What pattern did they draw?

Known quantities

Meaning in my words What do we mean by the known quantities in a problem?	**How can you tell?** A cart starts at rest and is timed for 3 s. Which two values can you write down?
Example or sketch A sensor reads +4 m at the start. How would you enter that known value?	**Non-example and why** A friend lists a guessed final speed among the given values. Where should it go?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Displacement

Meaning in my words How would you explain displacement to a friend?	**How can you tell?** Which two marks tell you the displacement even if the cart turns around?
Example or sketch Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement?	**Non-example and why** Someone adds both legs and gets 6 m. What did they find?

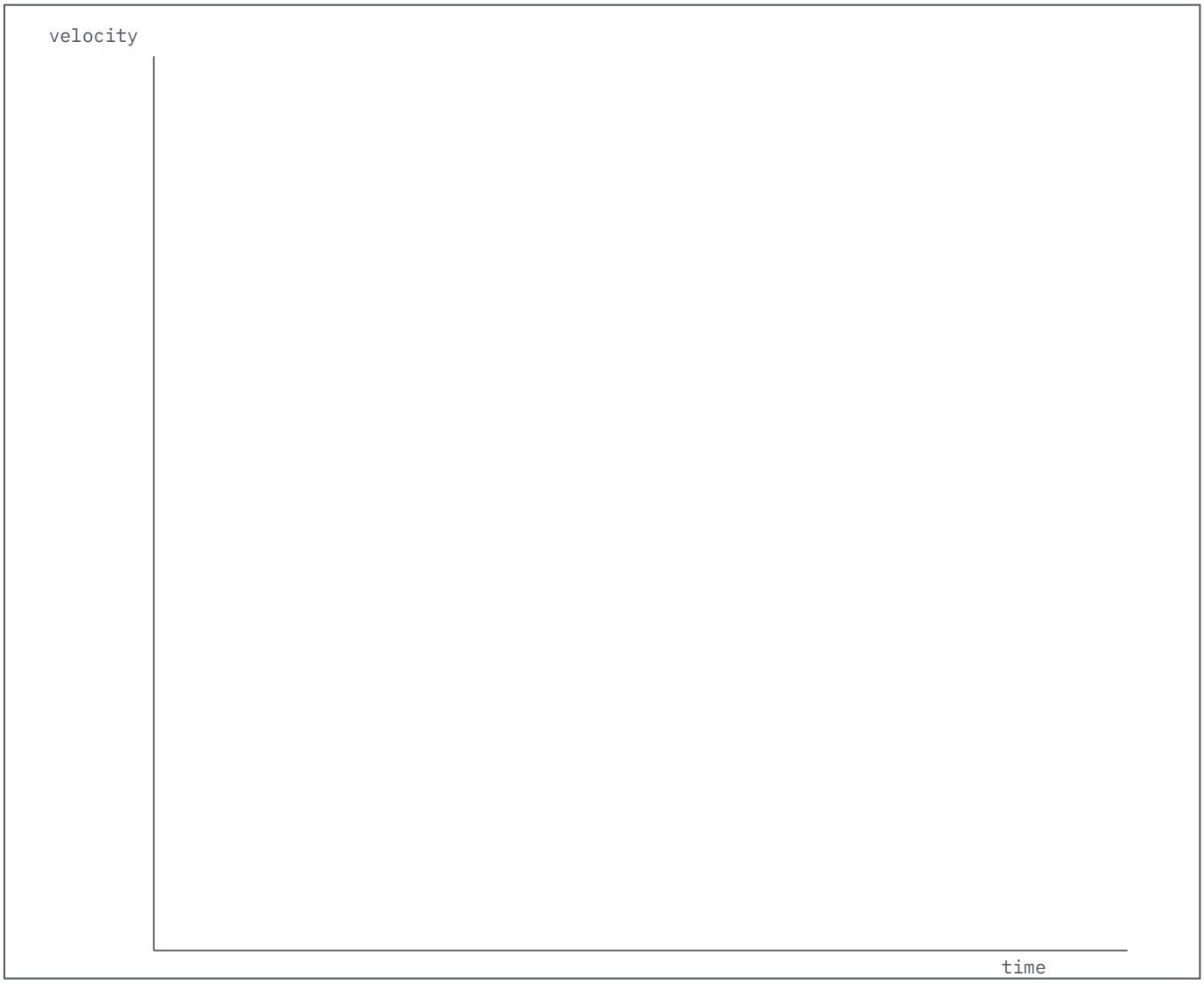
Cross-check

Meaning in my words How would you explain cross-check to a friend?	**How can you tell?** You found displacement using an equation. What feature of a velocity–time graph could check it?
Example or sketch A cart stays at +3 m/s for 2 s. How can a graph check the result of 6 m?	**Non-example and why** Someone copies the same number twice. What could actually catch a wrong answer?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a straight velocity–time line. Label which feature gives acceleration and which gives displacement.



My marks mean

Catch the next shot

08

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity-time graph to predict displacement.

CHOOSE ONE JOB TOGETHER

CAMERA ENGINEER ☐

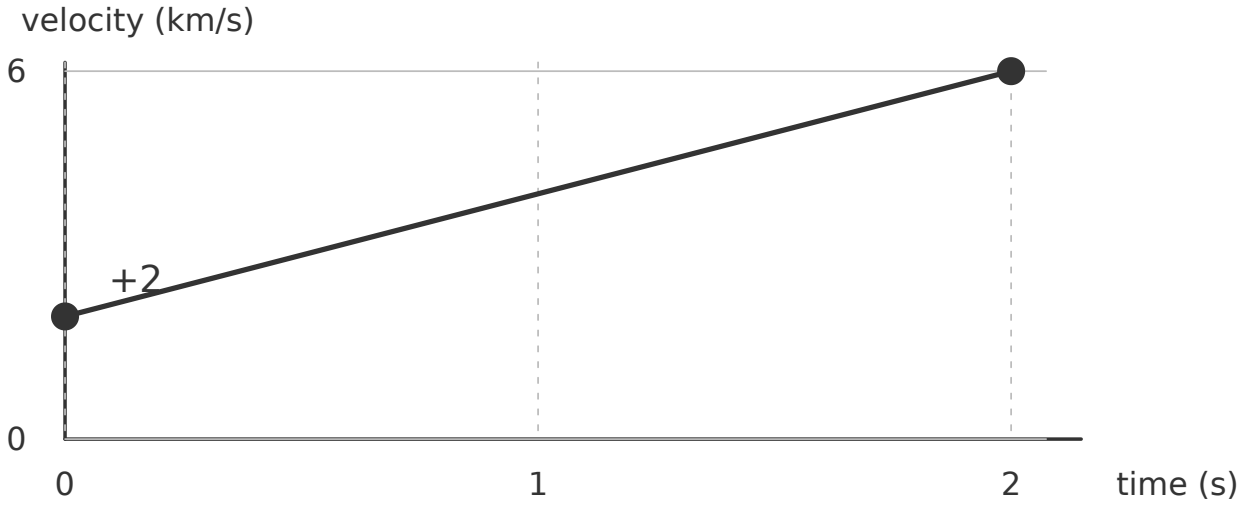
Shade the area under the velocity-time trace. Send the camera's displacement at the end of the shot.

REPLAY EDITOR ☐

Explain why using the final speed for the whole clip would miss the shot. Point to the graph as you speak.

Point to a clue. Keep your first marks; add the update using the color key.

XJ moves from **+2** to **+6 km/s** over **2 s**. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?



01 OUR FIRST CALL

Where should the camera look relative to the start? Show how the graph helps.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new report still let you guarantee that displacement? Explain your revised call.

Switch speakers. Check your first answer with a kinematics equation. What does that equation treat as constant?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks